

RESEARCH ARTICLE

Detection of H275Y oseltamivir resistance gene mutation among Influenza A(H1N1)pdm09 patients by allelic discrimination real-time RT-PCR

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Abstract

Influenza viruses can mutate genetically and cause a range of respiratory ailments. The H275Y mutation in the neuraminidase (NA) gene reduces the effectiveness of oseltamivir, a widely used drug for the treatment of Influenza A and B virus infection. The World Health Organization (WHO) recommends single-nucleotide polymorphism assays to detect this mutation. The present study aims to estimate the prevalence of H275Y mutation conferring oseltamivir resistance in Influenza A (H1N1)pdm09 virus among hospitalized patients from June 2014 to December 2021. Following the WHO protocol, allelic discrimination real-time RT-PCR was performed for 752 samples. Out of the 752 samples, 1 tested positive for Y275 gene mutation by allelic discrimination real-time RT-PCR. In samples of years 2020 and 2021, neither the H275 nor Y275 genotype was detected. Sequencing of the NA gene of all negative samples showed a mismatch between the NA sequence and the probes used in the allelic discrimination assay. Also, Y275 mutation was detected in only 1 sample from 2020. The prevalence of oseltamivir resistance was estimated as 0.27% among the Influenza A(H1N1)pdm09 patients during 2014–2021. The study highlights that the WHO-recommended probes for detecting H275Y mutation may not be useful to detect 2020 and 2021 circulating strains of Influenza A(H1N1)pdm09, emphasizing the need for continuous monitoring of mutations in the influenza virus.

KEYWORDS

allelic discrimination real-time RT-PCR, H275Y, Influenza A(H1N1)pdm09 virus, oseltamivir resistance

Smriti Krishna and Anup Jayaram are first authors.

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1 | INTRODUCTION

The influenza virus causes mild to severe respiratory infections resulting in 290 000–650 000 deaths globally.¹ The Influenza viruses are spherical, filamentous, or pleomorphic with a diameter of ~100 nm. They have a lipid envelope with characteristic spikes called haemagglutinin and neuraminidase (NA).² The fundamental activity of the NA protein involves cleaving of the sialic acids on the cell surface that cause the release of virions and infection of new cells.³ Oseltamivir carboxylate mimics the natural substrate of NA (sialic acid) and interacts with the active site, making Glu276 to rotate and bond with Arg224 which is stabilized by His275. These conformational alterations prevent the binding of sialic acid, and the subsequent release of the virus.⁴ The active sites of NA are known to be conserved, although mutations can affect these sites which may change virus transmissibility as well as drug sensitivity.^{5,6} A point mutation in NA gene, H275Y, which is due to a nucleotide change of Cysteine (C) to Thymine (T) at 823 position resulting in an amino acid change of Histidine (H) to Tyrosine (Y) at 275 position conferring oseltamivir resistance is one of the most significant mutations.^{5,7,8}

Antiviral usage has increased in recent times due to its effectiveness in reducing infections resulting in mild to moderate levels of resistance. Global Influenza Surveillance and Response System (GISRS) has deployed surveillance for influenza antiviral susceptibility testing. The GISRS states that phenotypic and genotypic assays are available to determine if an influenza virus is susceptible to NAIs. Genotypic assays comprise single-nucleotide polymorphism (SNP) detection, gene sequencing, and pyrosequencing.⁹ According to World Health Organization (WHO), the SNP based assay is currently used only for detection of H275Y mutation in influenza A(H1N1)pdm09 viruses and has been used in similar studies.^{10–14}

Resistance to oseltamivir for A/(H1N1)pdm09 started to appear and was reported to affect immunocompromised individuals¹⁵ as well as patients who were treated with the drug.⁵ Resistance to oseltamivir has been detected as 2.14% in China,¹⁶ 1.3% in Japan,¹⁷ 1.2% in United States,¹⁸ 1% in United Kingdom,¹⁵ and 0.74% in Indonesia.¹⁰ According to a study conducted by Trivedi et al.¹² in India, the prevalence rate was 0.15% while Saha et al.¹⁹ and Nandhini et al.²⁰ found that all the samples were sensitive to oseltamivir. In India, studies on the prevalence of oseltamivir resistant Influenza A (H1N1)pdm09 viruses are limited especially after the year 2015. Further, number of samples used in the previous studies are less and hence the possibility of detection of resistant virus is low. Thus, this study aims to detect the presence of H275Y mutation among Influenza A(H1N1)pdm09 patients from 2014 to 2021.

2 | MATERIALS AND METHODS

2.1 | Study population

Influenza A(H1N1)pdm09 positive samples archived at the Manipal Institute of Virology (MIV), from June 2014 to December 2021 were included in the study. Prevalence of oseltamivir resistance varies

from 0% to 2% in India.^{12,13} The sample size was estimated as 752, considering 2% prevalence ($p = 0.02$), 1% allowable margin of error ($P/2 = 0.01$ for rare disease), and 95% confidence interval using $n = \frac{z^2 P(1-P)}{d^2}$ where, n is the number of samples, z is the statistic corresponding to the level of confidence (1.96).²¹

Out of 3023 influenza A(H1N1)pdm09 positive samples archived at MIV from 2014 to 2021, a total of 752 samples with C_t values <30 were selected from each year from different states of India by random sampling.

2.2 | Allelic discrimination real-time RT-PCR

Influenza A(H1N1)-positive samples were tested used pooled sample testing strategy. Five positive samples with C_t value <30 were added to each pool making a total of 151 pools. Based on the sample volume required for manual extraction (140 μ L), 30 μ L of each sample was added to the pool and final elution volume was 60 μ L. Nucleic acid was extracted from these pools using FavorPrep™ Viral DNA/RNA kit (Favorgen Biotech Corporation). The extracted nucleic acids were subjected to allelic discrimination real-time RT-PCR as per the WHO protocol⁷ using QuantiTect Virus Kit (Qiagen). NA-specific primers and probes for Influenza A(H1N1)pdm09 as recommended by WHO⁷ were purchased from Applied Biosystems, and reconstituted. Positive controls used for wild type—H275 (Cat No. FRS1168) and mutant type—H275Y (Cat No. FRS1169) were standard strains procured from International Reagent Resource.

Assay components and reaction conditions were standardized for allelic discrimination real-time RT-PCR. Briefly, reaction mix contained 5 μ L of 5x QuantiTect Virus Master Mix, 0.5 μ L of both forward and reverse primers (2 μ M), 0.5 μ L of H275 and Y275 probe each of 2 μ M, 0.2 μ L of QuantiTect Virus RT Mix, 12.8 μ L of RNase free water, and 5 μ L of template, making a total of 25 μ L. Cycling conditions were programmed as a pre-read stage of 60°C for 1 min, reverse transcription process at 50°C for 20 min, followed by initial denaturation at 95°C for 10 min, with subsequent 45 cycles of amplification (denaturation at 95°C for 15 s and annealing and extension at 58°C for 45 s), and lastly, a post-read stage of 60°C for 1 min. PCR was performed using QuantStudio™ 6 Flex Real-Time PCR System (Applied Biosystems).

2.3 | Modified Sanger sequencing

Sample homozygous for Y275 or negative for both H275 and Y275 targets in the allelic discrimination real-time RT-PCR were selected for sequencing. The partial fragment of NA gene was amplified using WHO-recommended primers. The desired product of 620 bp was extracted using GenElute™ Gel Extraction Kit (Sigma-Aldrich). Sequencing was done using BigDye Terminator (version 3.1) cycle sequencing kit (Applied Biosystems) and analyzed using 3500xL Genetic analyzer (Applied Biosystems) following the manufacturer's instructions.

2.4 | Sequence analysis

Sequence analysis was done using MEGA X version 11 software. The reference sequence used for the analysis of 2014–2017 samples, Influenza A/Michigan/10/2009(H1N1)-like virus, was downloaded from National Center for Biotechnology Information. Reference sequences for 2020 and 2021 samples including WHO recommended vaccine strains sequences for Influenza A/Hawaii/70/2019(H1N1)-like virus and Influenza A/Wisconsin/588/2019(H1N1)-like viruses were downloaded from Global Initiative on Sharing All Influenza Data (GISAID).

3 | RESULTS

3.1 | Study population

Among the 752 samples selected from the different states of India for detection of H275Y gene mutation, the highest number of samples tested were from the state of Karnataka (40.4%, $n = 304$) followed by Tamil Nadu (19.6%, $n = 148$). The 2017 influenza season had the highest number of influenza cases hence a greater number of samples were selected for testing. The year-wise and geographic details of samples used in the study are summarized in Table 1.

3.2 | Allelic discrimination real-time RT-PCR

Endpoint fluorescent signals were analyzed and the threshold was adjusted manually. The results were categorized as homozygous wild-type (H275), homozygous mutant-type (Y275), and heterozygous type (H275/Y275).

Out of the 151 pools that were tested for H275Y mutation, 140 pools were homozygous H275 and only 1 pool showed amplification for both H275 and Y275 (heterozygous type) and 10 pools were negative for both H275 and Y275 (Figure 1 and Table 1).

The five samples of the heterozygous positive pool were tested individually and one sample was detected as homozygous mutant-type (Y275). The prevalence rate of oseltamivir resistance from 2014 to 2019 was found to be 0.13%.

3.3 | Modified Sanger sequencing

Sequence analysis of homozygous Y275 sample in comparison to reference strains showed a nucleotide change of CAC to TAC at 823 position resulting in an amino acid change from H to Y at 275 position, confirming the presence of H275Y mutation.

Fifty samples from the 10 undetermined pools of 2020 and 2021 were sequenced and sequence analysis showed no nucleotide change at 823 position for 49 samples, however one sample from 2020 showed nucleotide change of CAC to TAC at 823 position resulting in an amino acid change from H to Y at 275 position. The prevalence rate of oseltamivir resistance from 2014 to 2021 was found to be

0.27%. All the nucleotide sequences (510 bp) from this study were deposited in GenBank with accession numbers from OP737518 to OP737538 and OQ363464 to OQ363489. Among these, OP737518 and OQ363483 had the codon change of CAC to TAC with amino acid mutation of H to Y.

Sequence alignment of all 2020 and 2021 samples with the reference sequences and WHO recommended probe showed a mutation of GAG to GAA at 831 position. This nucleotide mismatch with probe could have possibly resulted in negative results for both H275 and Y275 targets in allelic discrimination PCR.

Further, nucleotide BLAST analysis of the probe binding region using GISAID analysis tool showed that the nucleotide change of G to A at 831 position has been reported since end of 2019 (GISAID No. EPI_ISL_16071680) in different regions of the world and this nucleotide change is also observed in the WHO recommended vaccine strain for 2021–2022–Northern hemisphere (Figure 2).

4 | DISCUSSION

Oseltamivir is the most widely used antiviral for the treatment of influenza infections in India, therefore it is important to evaluate its susceptibility and effectiveness. H275Y mutation leading to oseltamivir resistance has a prevalence of 1%–2% worldwide and less than 1% in India. Gohil et al.²² conducted a study for the surveillance of the influenza virus in Mumbai and found that all the samples were sensitive to oseltamivir. Similar studies by Sahu et al.¹³ during 2009–2015 and Trivedi et al.¹² during 2015–2016 demonstrate the prevalence of oseltamivir resistance as 1.4% and 0.15%, respectively.

In India, the number of published studies on oseltamivir resistance after the year 2015 are sparse. Hence, this study was conducted to estimate the prevalence of oseltamivir resistance among Influenza A (H1N1)pdm09 patients from 2014 to 2021 using allelic discrimination real-time RT-PCR to identify H275Y gene mutation.

The current study, estimates prevalence rate of oseltamivir resistance as 0.27% among hospitalized patients that was found to be similar to previous studies from India.^{12–14,19,20} Based on the results, it can be concluded that oseltamivir resistance is still less than 1% in India. Out of two resistant samples, one was from the year 2017, collected from 26-year-old male, with 2 days history of fever, cough, coryza, red eye, retro-orbital pain, myalgia, and mild gastrointestinal symptoms. He had received oseltamivir treatment and there was no history of previous influenza infection or oseltamivir treatment. However, the follow-up report showed the patient had recovered and was healthy. The second resistant sample (collected in 2020) was from a 55-year old male, with 2 days history of fever, cough, difficulty in swallowing, breathlessness, and gastrointestinal symptoms. The second patient had also received oseltamivir treatment with no previous history of infection or resistance. However, the patient's condition deteriorated and hypoxia worsened, secondary bacterial infection was also detected. Due to the patient's financial restraints he was discharged against medical advice. A few studies from Hong Kong,²³ Vietnam,²⁴ and United States,¹⁸ have reported oseltamivir resistance in patients without prior history of

TABLE 1 Year-wise geographic distribution of clinical samples and results of allelic discrimination PCR.

SI No	Year	State	Number of samples	Number of pools	Allelic discrimination real time RT-PCR results			
					Homozygous H275	Homozygous Y275	Heterozygous H275/Y275	Undetermined
1	2014	Karnataka	15	3	3	0	0	0
2	2014 & 2015	Karnataka	5	1	1	0	0	0
3	2015	Karnataka	23	6	6	0	0	0
		Kerala	7					
4	2015 & 2016	Kerala	1	1	1	0	0	0
		Assam	3					
		Gujarat	1					
5	2016	Karnataka	15	4	4	0	0	0
		Kerala	1					
		Tamil Nadu	3					
		Tripura	1					
6	2017	Assam	22	51	50	0	1	0
		Goa	10					
		Gujarat	33					
		Jharkhand	19					
		Karnataka	46					
		Kerala	26					
		Maharashtra	27					
		Odisha	25					
		Tamil Nadu	47					
7	2017 & 2018	Tamil Nadu	4	1	1	0	0	0
		Assam	1					
8	2018	Assam	2	35	35	0	0	0
		Gujarat	6					
		Karnataka	56					
		Kerala	23					
		Maharashtra	9					
		Tamil Nadu	79					
9	2018 & 2019	Tamil Nadu	2	1	1	0	0	0
		Odisha	2					
		Assam	1					
10	2019	Assam	3	38	38	0	0	0
		Goa	4					
		Gujarat	8					
		Jharkhand	26					
		Karnataka	96					
		Kerala	18					
		Maharashtra	15					
		Odisha	4					
		Tamil Nadu	13					

TABLE 1 (Continued)

SI No	Year	State	Number of samples	Number of pools	Allelic discrimination real time RT-PCR results			
					Homozygous H275	Homozygous Y275	Heterozygous H275/Y275	Undetermined
11	2020	Karnataka	35	7	0	0	0	7
12	2020 & 2021	Goa	1	1	0	0	0	1
		Karnataka	3					
		Kerala	1					
13	2021	Karnataka	10	2	0	0	0	2
		Total	752	151	140	0	1	10

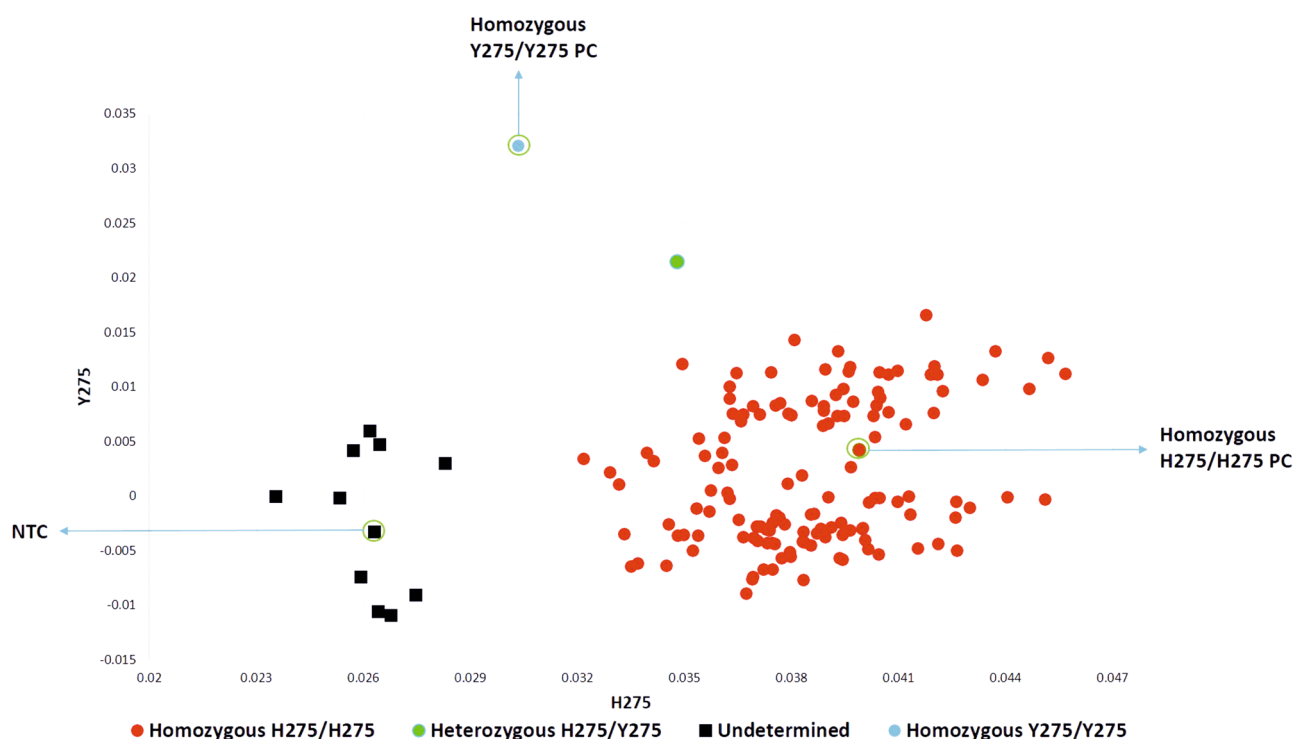


FIGURE 1 Result of allelic discrimination real-time RT-PCR. The red dots toward x-axis represent the pools that are homozygous H275, including the positive control shown encircled. The blue dots toward y-axis represent the positive control for homozygous Y275. The green dot represents the pool which was heterozygous H275/Y275. The black box represents the undetermined pools, including negative control shown encircled.

treatment. A similar finding was reported by Tandel et al.²⁵ in a study conducted in Madhya Pradesh, India, where drug resistance was detected in patients without any previous exposure. Oseltamivir resistant and oseltamivir-susceptible viral infections are reported to be clinically similar, with the exception of those who are immunocompromised or have comorbid conditions.²⁶

Allelic discrimination real-time RT-PCR is a rapid SNP detection method that detects H275Y mutation and has been recommended by WHO. In the current study, pooled sample testing strategy was used to detect H275Y gene mutation. Pooled sample testing strategy has been used to detect viruses such as hepatitis B virus, human immunodeficiency virus, influenza virus among others.²⁷ This testing strategy

with pool sizes of 5–10 samples has also been widely used to increase testing capacity and conserve resources during upsurge testing periods such as the on-going COVID-19 pandemic.²⁸ A study by Shukla et al. showed that the average change in C_t for pool of 5 was +2.56 and +3.38 for pool of 10.²⁸ To avoid false positives and increase efficiency of allelic discrimination real-time RT-PCR, pools of 5 influenza A(H1N1)pdm09 positive samples with $C_t < 30$ were tested for SNP detection.

In this study, the samples of 2020 and 2021 were negative for both wild-type and mutant-type. Based on sequencing results, it was confirmed that 49 samples did not have the H275Y mutation, but one sample was detected with H275Y mutation. On further analysis of sequences, an additional nucleotide mutation was detected at the

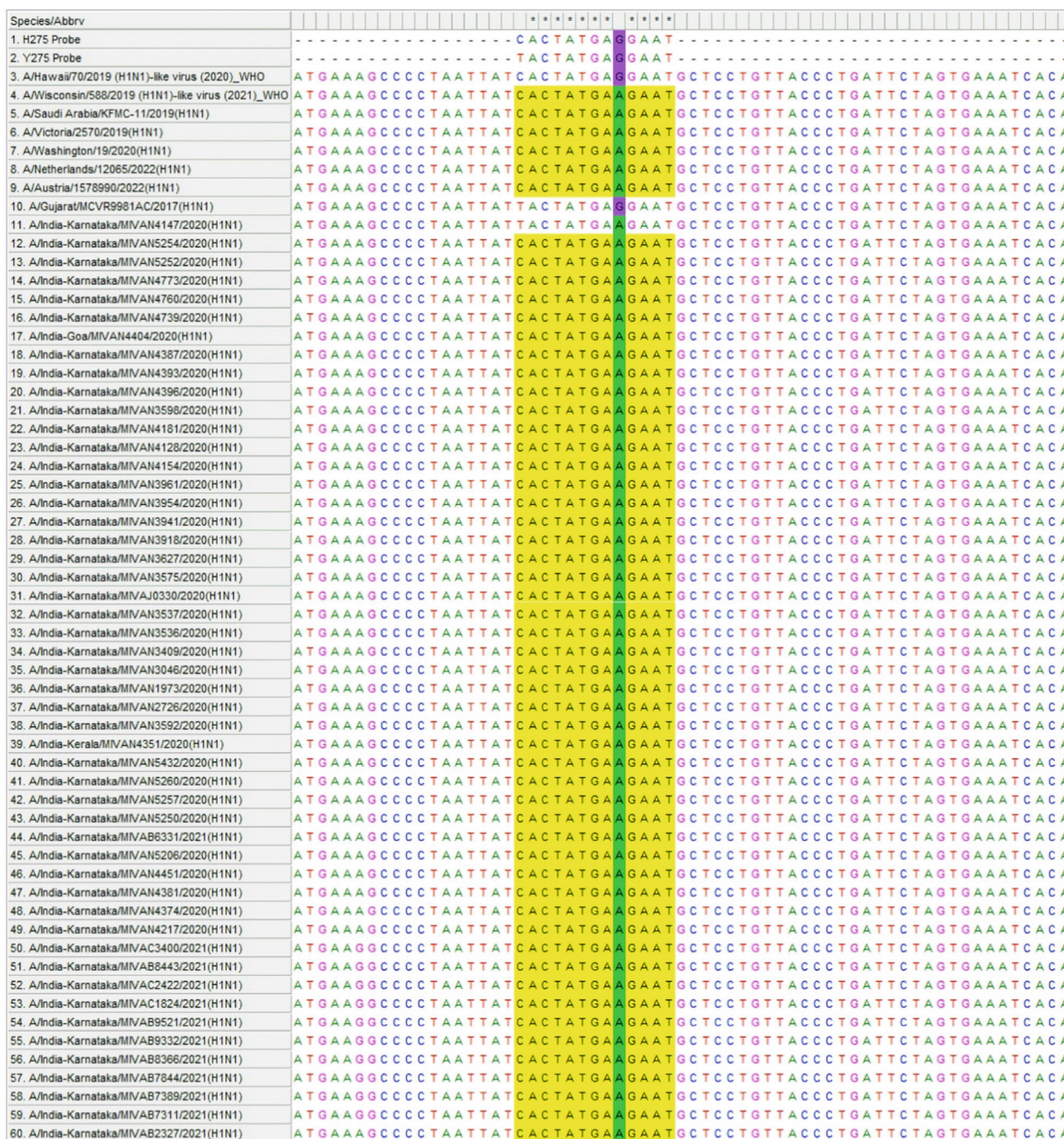


FIGURE 2 Sequence alignment of the 2017, 2020, and 2021 samples with H275 probe, Y275 probe, World Health Organization recommended vaccine strain (No. 3 & 4) and reference sequences (No. 5–9). The yellow highlighted region is the probe binding site and green highlighted region shows a nucleotide change of GAG to GAA at 831 position.

probe binding site that might have contributed to negative results for both H275 and Y275. Allelic discrimination assays are very specific and a single mismatch between the template and the probe sequence will affect its hybridization.²⁹

Further, nucleotide BLAST analysis of probe binding region showed that these mutations might have emerged toward the end of 2019 (GISAID No. EPI_ISL_16071680, unpublished data) and has been detected in many countries since its emergence.

The WHO recommended vaccine strains sequence for Northern Hemisphere—2021–2022 also documents this change (GISAID No. EPI_ISL_404527). Although, the nucleotide change at 831 position results in a synonymous mutation and does not affect drug binding, the mismatch at the probe binding site is a major concern since these probes are recommended by WHO for detection of the substitution at amino acid 275 in the NA of influenza A(H1N1)pdm09 viruses. The WHO, allelic discrimination real-time RT-PCR is considered as a rapid

and convenient method and is widely used for screening of mutations associated with drug resistance.¹² The detection of nucleotide change in probe binding site of 2020 and 2021 samples strengthens the fact that considering the high mutation rates among influenza A viruses the WHO protocol recommendations needs to be revised periodically to avoid false results.

5 | CONCLUSION

The evolving nature of influenza viruses poses a challenge for developing vaccines and antivirals. The NA inhibitors against influenza viruses are effective and used extensively. Any mutation in the drug's binding site could result in a decrease in the drug's effectiveness and eventual discontinuation. This study illustrates a 7-year profile of H275Y gene mutation leading to oseltamivir resistance among Influenza A(H1N1)pdm09 patients from 2014 to 2021. The result shows a lower prevalence of 0.27% which indicates that oseltamivir carboxylate can be continued to be used as a therapeutic. The negative results obtained in the samples of 2020 and 2021 using allelic discrimination real-time RT-PCR following WHO protocol highlights the necessity of monitoring the mutations in the influenza virus and implementing it in the respective assays so as to avoid any false negative results.

AUTHOR CONTRIBUTIONS

Anitha Jagadesh conceived, and guided the research work. Smriti Krishna performed most of the experiments. Anup Jayaram contributed to data processing and analysis. Ujwal Shetty contributed to performing sequencing. Prasad Varamballi contributed in data analysis. Chiranjay Mukhopadhyay provided insightful discussion. All authors have read and approved the manuscript

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data supporting the current study is available from the corresponding author upon request.

ETHICS STATEMENT

The study was reviewed and approved by the Institutional Ethical Committee (IEC), Kasturba Medical College, and Kasturba Hospital (Approval ID-IEC2: 179/2022).

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REFERENCES

- World Health Organization. Burden of disease. Accessed April 1, 2022. <https://www.who.int/teams/globalinfluenza-programme/surveillanceandmonitoring/burden-of-disease>
- Knipe DM, Howley PM, Griffin DE, et al. *Fields Virology, Volumes 1 and 2*. Wolters Kluwer.
- Cohen M, Zhang XQ, Senaati HP, et al. Influenza A penetrates host mucus by cleaving sialic acids with neuraminidase. *Virol J*. 2013; 10(1):321.
- Grayson ML. *Kucers' the Use of Antibiotics: a Clinical Review of Antibacterial, Antifungal, Antiparasitic and Antiviral Drugs*. 6th ed. Hodder Arnold; 2010.
- Escuret V, Ferraris O, Lina B. The antiviral resistance of influenza virus. *Therapy*. 2011;8(6):741-762.
- McAuley JL, Gilbertson BP, Trifkovic S, Brown LE, McKimm-Breschkin JL. Influenza virus neuraminidase structure and functions. *Front Microbiol*. 2019;10:39.
- WHO. Real-time RT-PCR allelic discrimination analysis for detection of the substitution at amino acid 275 in the neuraminidase (NA) of A (H1N1) pdm09 Influenza viruses. Influenza Virus Research Center National Institute of Infectious Diseases Tokyo, Japan, 1–6. Accessed March 28, 2022. https://cdn.who.int/media/docs/defaultsource/influenza/antiviralwgsurveillance/allelic_discrimination_analysis_na275.pdf?sfvrsn=4e616506_10
- Von Itzstein M. The war against influenza: discovery and development of sialidase inhibitors. *Nat Rev Drug Discovery*. 2007;6(12): 967-974.
- World Health Organization. *Practical Guidance for National Influenza Centres Establishing or Implementing Neuraminidase Inhibitor Susceptibility Surveillance*. World Health Organization; 2016:18. <https://apps.who.int/iris/handle/10665/205703>
- Pawestri HA, Nugraha AA, Hariastuti NI, Setiawaty V. Detection of neuraminidase inhibitor-resistant influenza A (H1N1)pdm09 viruses obtained from influenza surveillance in Indonesia. *SAGE Open Med*. 2018;6:205031211881829.
- Roosenhoff R, Schutten M, Reed V, et al. Secondary substitutions in the hemagglutinin and neuraminidase genes associated with neuraminidase inhibitor resistance are rare in the Influenza Resistance Information Study (IRIS). *Antiviral Res*. 2021;189: 105060.
- Trivedi K, Malhotra B, Dubey W, et al. Detection of H275Y mutation conferring oseltamivir drug resistance in influenza A (H1N1) pdm09 virus. *J Pure Applied Microbiol*. 2021;15(3):1421-1428.
- Sahu M, Singh N, Shukla MK, et al. Molecular and epidemiological analysis of pandemic and post-pandemic influenza A(H1N1)pdm09 virus from central India. *J Med Virol*. 2018;90(3):447-455.
- Potdar VA, Dakhane MR, Abraham A, et al. Antiviral drug profile of human influenza A & B viruses circulating: 2004-2011. *Indian J Med Res*. 2014;140(2):244-251.
- Lackenby A, Moran Gilad J, Pebody R, et al. Continued emergence and changing epidemiology of oseltamivir-resistant influenza A (H1N1)2009 virus, United Kingdom, winter 2010/11. *Euro Surveill*. 2011;16(5):1-6.
- Huang W, Li X, Cheng Y, et al. Characteristics of oseltamivir-resistant influenza A (H1N1) pdm09 virus during the 2013-2014 influenza season in Mainland China. *Virol J*. 2015;12(1):96.
- Chong Y, Matsumoto S, Kang D, Ikematsu H. Consecutive influenza surveillance of neuraminidase mutations and neuraminidase inhibitor resistance in Japan. *Influenza Other Respir Viruses*. 2019;13(2): 115-122.
- Okomo-Adhiambo M, Fry AM, Su S, et al. Oseltamivir-resistant influenza A(H1N1) pdm09 viruses, United States, 2013-14. *Emerging Infect Dis*. 2015;21(1):136-141.
- Saha P, Biswas M, Gupta R, et al. Molecular characterization of Influenza A pandemic H1N1 viruses circulating in eastern India

- during 2017–19: antigenic diversity in comparison to the vaccine strains. *Infect Genet Evol.* 2020;81:104270.
20. Nandhini P, Sistla S. Genetic sequencing of Influenza A (H1N1) pdm09 isolates from South India, collected between 2011 and 2015 to detect mutations affecting virulence and resistance to oseltamivir. *Indian J Med Microbiol.* 2020;38(3–4):324–337.
 21. Arya R, Antonisamy B, Kumar S. Sample size estimation in prevalence studies. *Indian J Pediatr.* 2012;79(11):1482–1488.
 22. Gohil D, Kothari S. Oseltamivir resistant influenza A (H1N1) virus infection in Mumbai, India. *J Antivirals Antiretrovirals.* 2015;7(4):108–114.
 23. Leung TWC, Tai ALS, Cheng PKC, Kong MSY, Lim W. Detection of an oseltamivir-resistant pandemic influenza A/H1N1 virus in Hong Kong. *J Clin Virol.* 2009;46(3):298–299.
 24. Guttler SD. The case for integrated delivery systems: a community cluster of oseltamivir-resistant cases of 2009 H1N1 influenza. *N Engl J Med.* 2010;362:86–87.
 25. Tandel K, Sharma S, Dash PK, Parida M. Oseltamivir-resistant influenza A(H1N1)pdm09 virus associated with high case fatality, India 2015. *J Med Virol.* 2018;90(5):836–843.
 26. Renaud C, Kuypers J, Englund JA. Emerging oseltamivir resistance in seasonal and pandemic influenza A/H1N1. *J Clin Virol.* 2011;52:70–78.
 27. Van TT, Miller J, Warshawer DM, et al. Pooling nasopharyngeal/throat swab specimens to increase testing capacity for influenza viruses by PCR. *J Clin Microbiol.* 2012;50(3):891–896.
 28. Shukla S, Upadhyay V, Maurya VK. Evaluating the efficiency of specimen (sample) pooling for real-time PCR based diagnosis of COVID-19. *Indian J Med Microbiol.* 2021;39(3):339–342.
 29. Heissl A, Arbeithuber B, Tiemann-Boege I. High-throughput genotyping with TaqMan allelic discrimination and allele-specific genotyping assays. In: White SJ, Cantsilieris S, eds. *Genotyping: Methods and Protocols.* Springer; 2017:29–57. doi:10.1007/978-1-4939-6442-0_3

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